

United States Tree Observations and Species Range Patterns

A focused GlobalGeoTree analysis with regional context

Research Focus

This report focuses on the **United States** while keeping a short regional context.

RQ3 - Regional and Country-Level Distribution

How are tree observations and recorded species distributed in the United States compared with major reference regions and countries?

RQ4 - Species Range and Category Patterns

Do observations of the same species, or species in the same Rare/Common/Frequent category, tend to cluster in similar geographic areas?

The report is intentionally selective. It uses fewer figures than the exploratory version so that each visualization has a clear role in the argument.

Data Loading

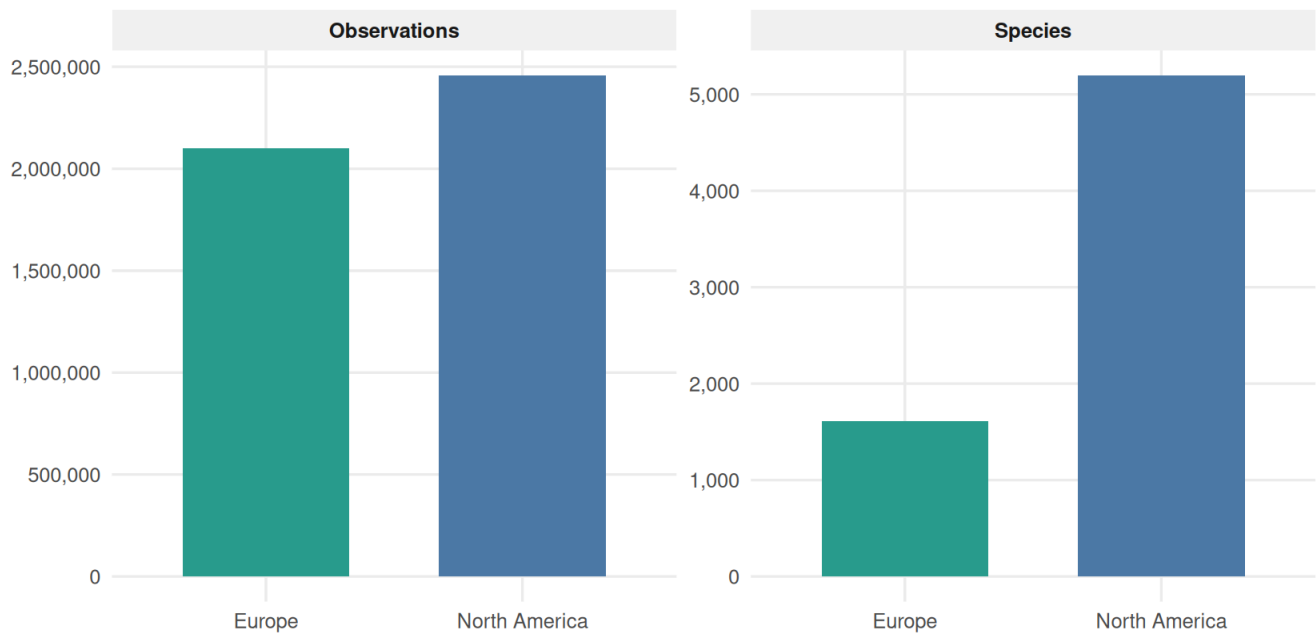
Metric	Value
US observations	1,932,465
US recorded species	3,001
US recorded genera	669
US recorded families	156
US observations with known category	711,966
US species with known category	390

Category source: **GlobalGeoTree-10kEval-900.csv** joined by **species_key**.

Regional Context

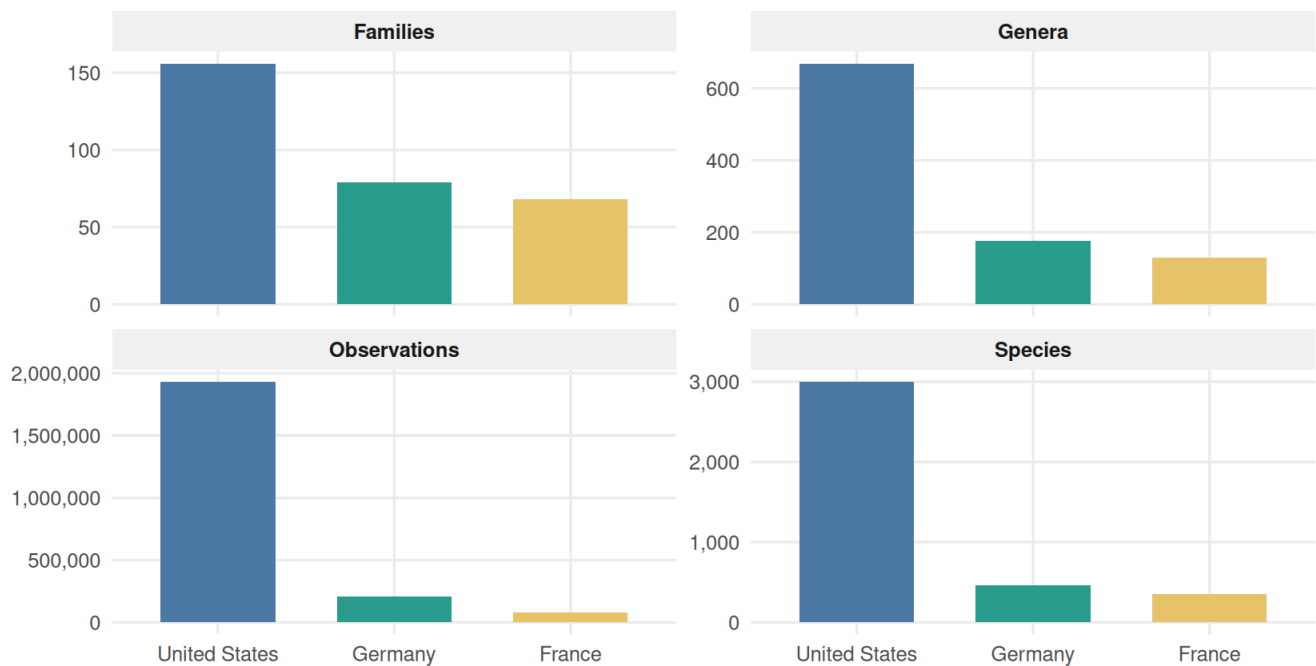
North America and Europe as regional context

North America is strongly influenced by the large US contribution in GlobalGeoTree.



United States, Germany, and France comparison

The US contributes far more observations and recorded taxa in this dataset.

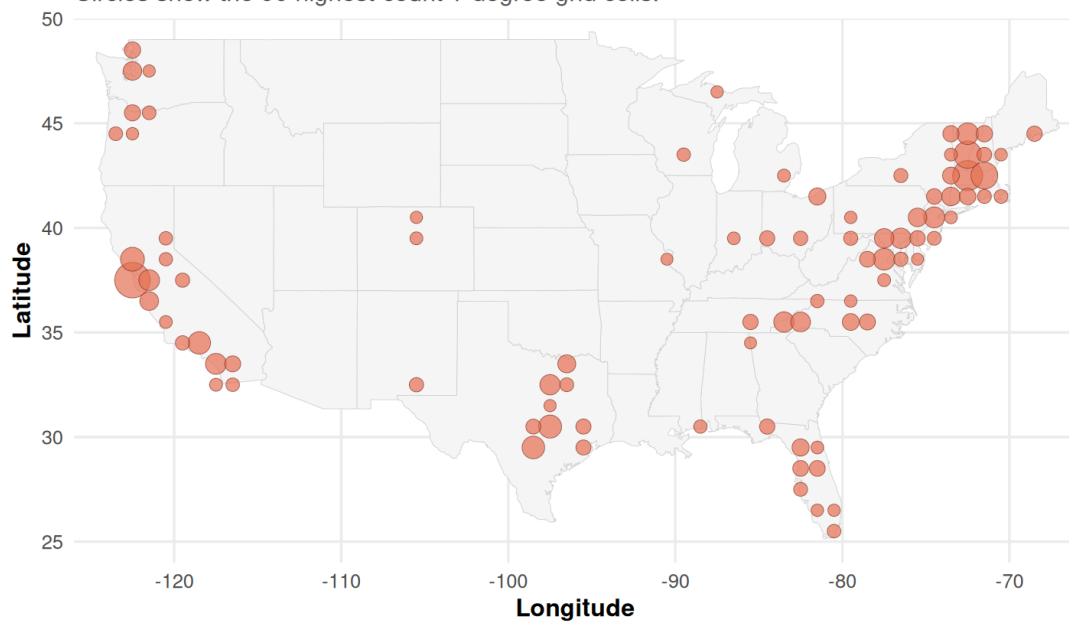


Interpretation. The United States is the main driver of the North American subset. Comparisons with Germany and France should therefore be interpreted as comparisons of dataset coverage and recording intensity, not only as biological richness.

United States Hotspots

Largest US observation hotspot areas

Circles show the 90 highest-count 1-degree grid cells.

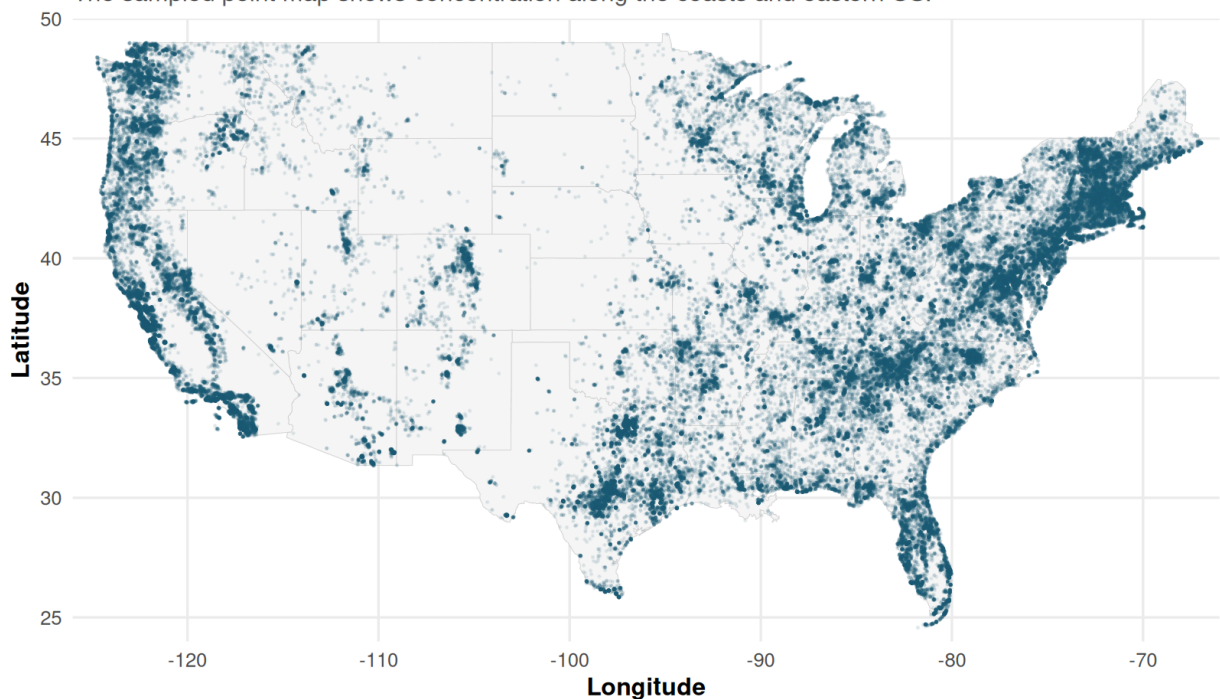


Records ● 10,000 ● 20,000 ● 30,000 ● 40,000 ● 50,000

Hotspot circles summarize observation concentration without using rectangular heatmap cells.

Spatial footprint of US observations

The sampled point map shows concentration along the coasts and eastern US.



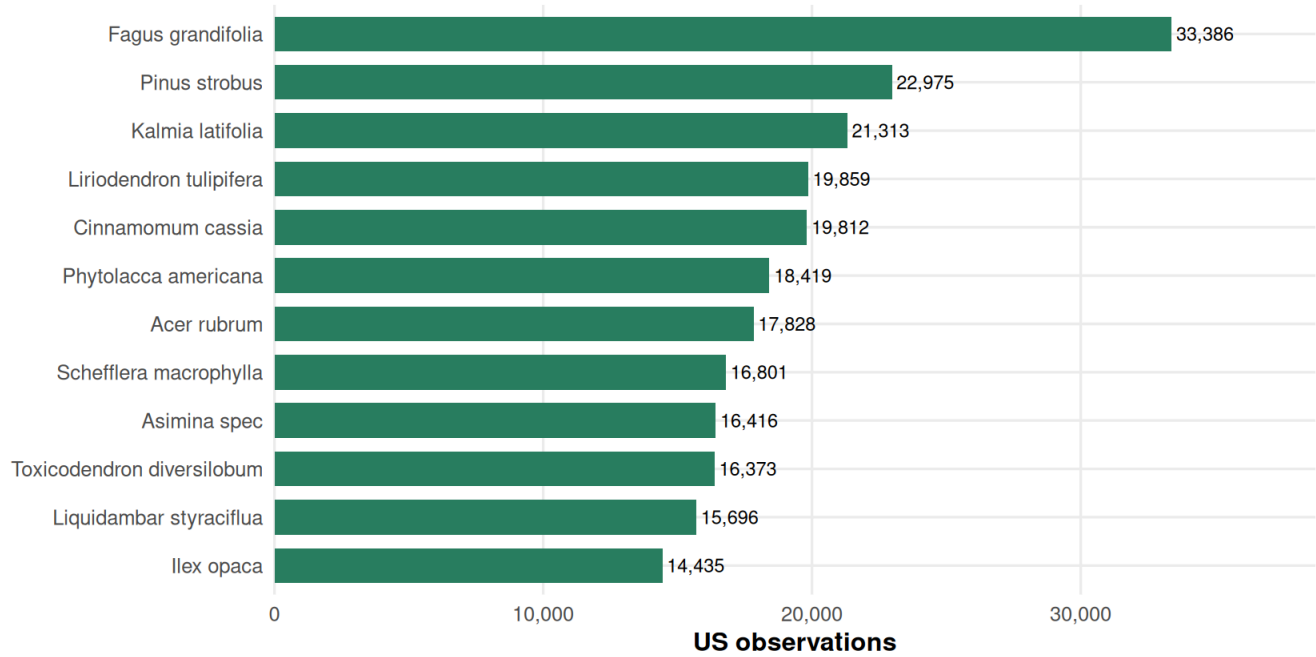
Points are sampled for readability.

Interpretation. US observations are not evenly distributed. The strongest concentrations are visible in the eastern US, West Coast, Florida, the Pacific Northwest, and parts of the Great Lakes region. Sparse areas may represent fewer observations, lower platform use, lower population density, access limitations, or genuinely different tree distributions.

Most Recorded US Species

Most recorded US taxa in GlobalGeoTree

High observation counts may reflect popularity, detectability, and platform coverage.



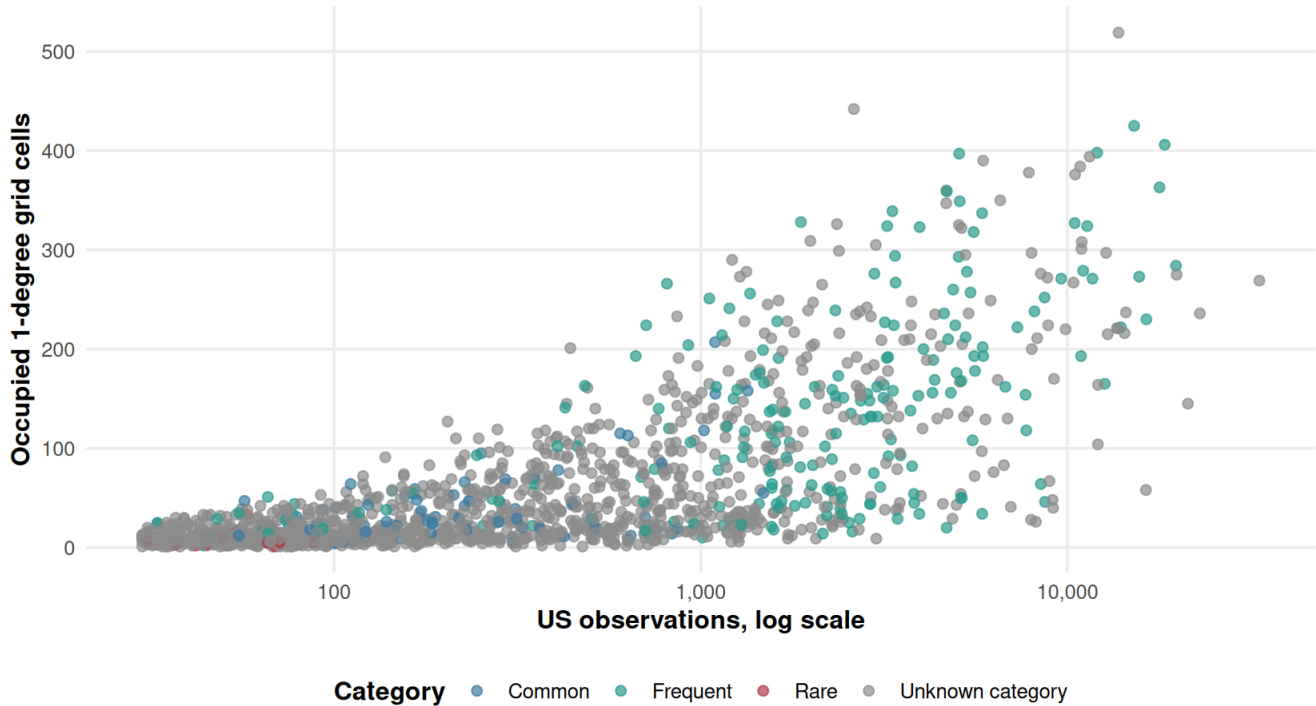
This chart uses the dataset's species labels as supplied; it does not reclassify tree status.

Interpretation. The most recorded taxa are not necessarily the most abundant trees in the United States. They are the most recorded taxa in this dataset.

Species Range Patterns

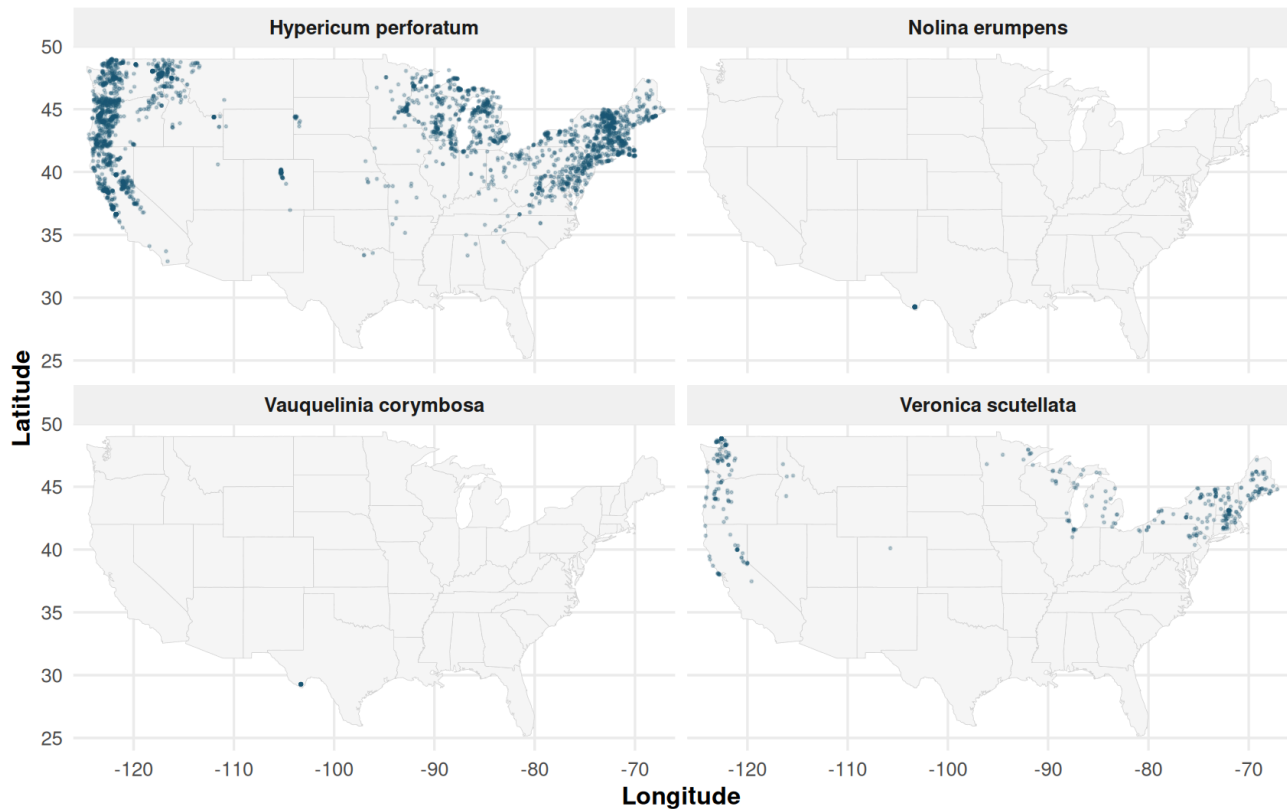
US species differ strongly in spatial spread

Each point is one species with enough observations for spread analysis.



Examples of localized and widespread US species

Selected species show low and high median distance to their species center.

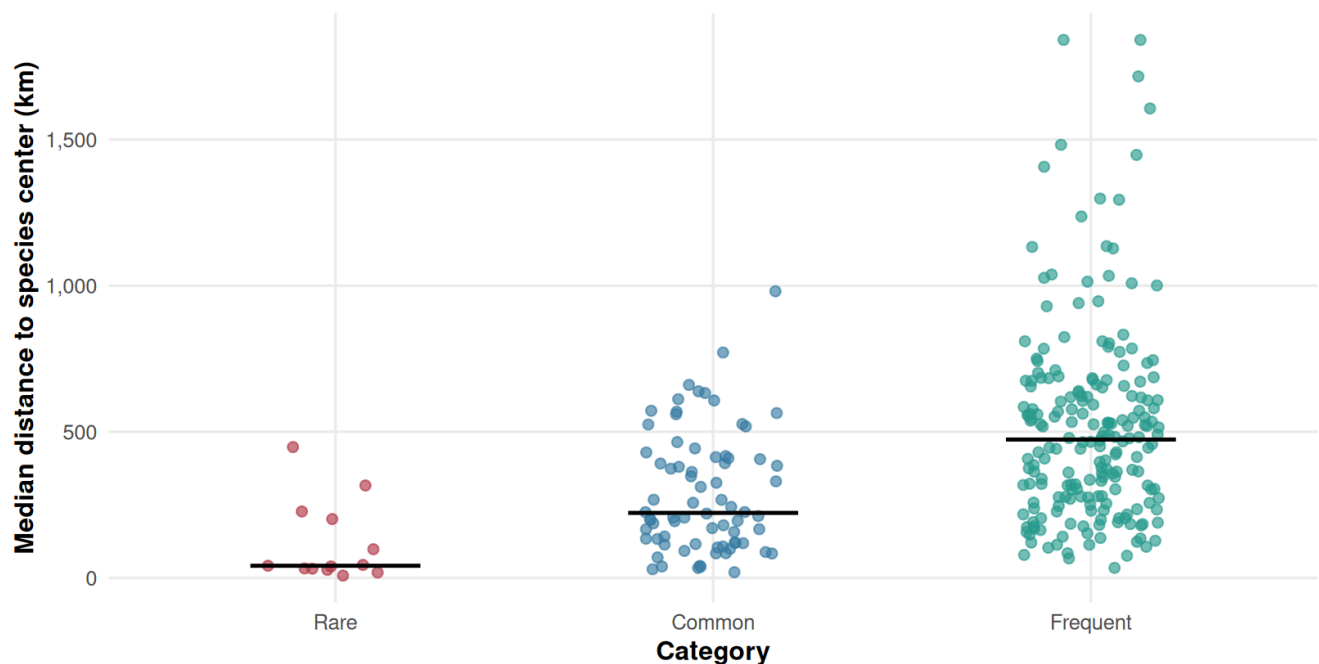


Interpretation. Some species occupy many spatial grid cells, while others are tightly localized. However, a localized pattern can reflect true range limits, low observation effort, or the way a species entered GlobalGeoTree.

Rare, Common, and Frequent Categories

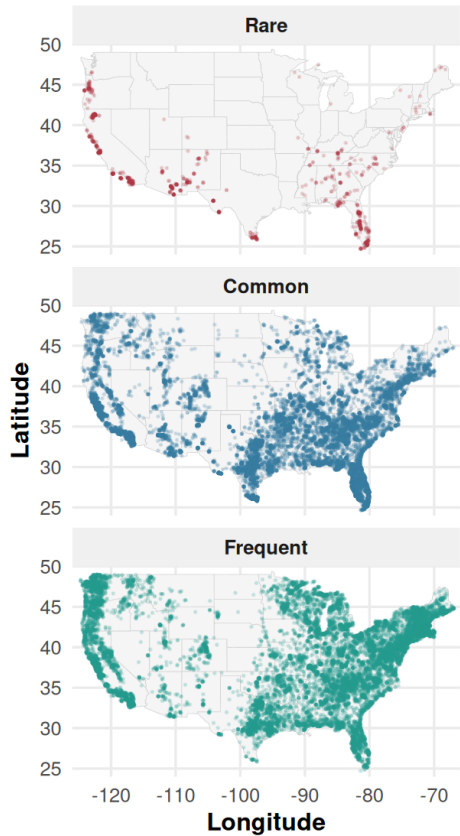
Spatial spread differs by category

Each point is one categorized species; black bars show medians.



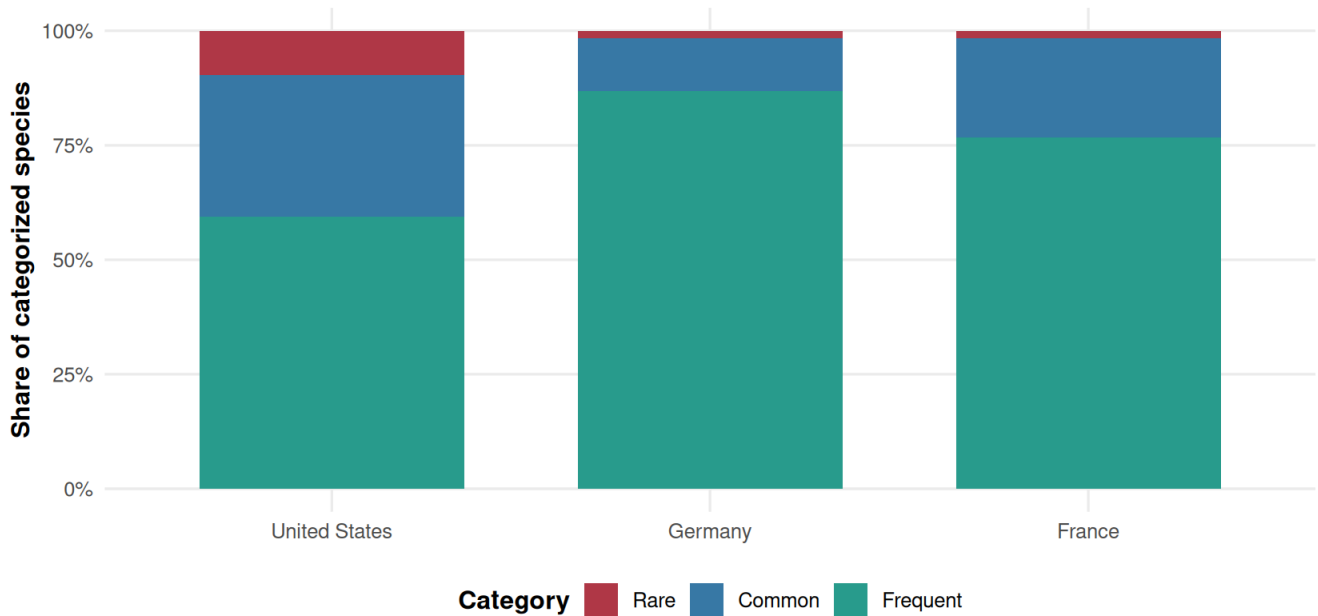
US locations by species category

Mapped only for observations whose species matched Rare/Common/Freque



Category composition across the US, Germany, and France

Bars show the category composition of species with known category labels.



Interpretation. Category results apply only to species matched to a category label. In this run, the US has **711,966** observations and **390** species with known categories. The category analysis should therefore be treated as a categorized subset, not the full US dataset.

Overall Interpretation

The United States is the dominant part of the North American subset in GlobalGeoTree. Within the US, observations are spatially concentrated rather than evenly distributed, with strong hotspots along the coasts and in the eastern half of the country. Species also differ strongly in spatial spread: some are widespread across many grid cells, while others appear localized. Category patterns suggest that Frequent species tend to be more spatially widespread than Rare species, but this conclusion depends on the matched category subset and should not be generalized to all US species without checking category coverage.